ECON 1550: International Finance

National Income Accounting for Open Economies

Announcements

- Problem Set 1 due today before midnight
- Solutions and Problem Set 2 posted right after
- Read pages 58–67 of textbook before FRIDAY ~~Wednesday~~ lecture
- Fill in survey for office hours and section (right now!)

Office Hours and Section Availability

Please fill in your availability:

https://www.when2meet.com/?34820140-APJfJ

Agenda

- Balance of payments account
- Exchange rate determination with asset approach

Macro Review: GDP

Gross Domestic Product (GDP) measures the total value of:

- **Production:** All final goods and services produced within a country
- **Income:** All income earned from production within a country
- **Value added:** Sum of value added at each stage of production

All three approaches yield the same number.

Closed Economy

$$Y = C + I + G$$

- All output is either consumed, invested, or purchased by government
- No trade with the rest of the world

Open Economy with GDP

$$GDP = C + I + G + \underbrace{EX - IM}_{NX}$$

- EX = Exports (domestic goods sold abroad)
- IM = Imports (foreign goods purchased domestically)
- NX = Net exports (trade balance)

GDP and GNP

- **GDP:** Value of production *within a country's borders*
  - Regardless of who owns the factors of production
- **GNP:** Value of production by *a country's residents*
  - Regardless of where production takes place
- **Relationship:**

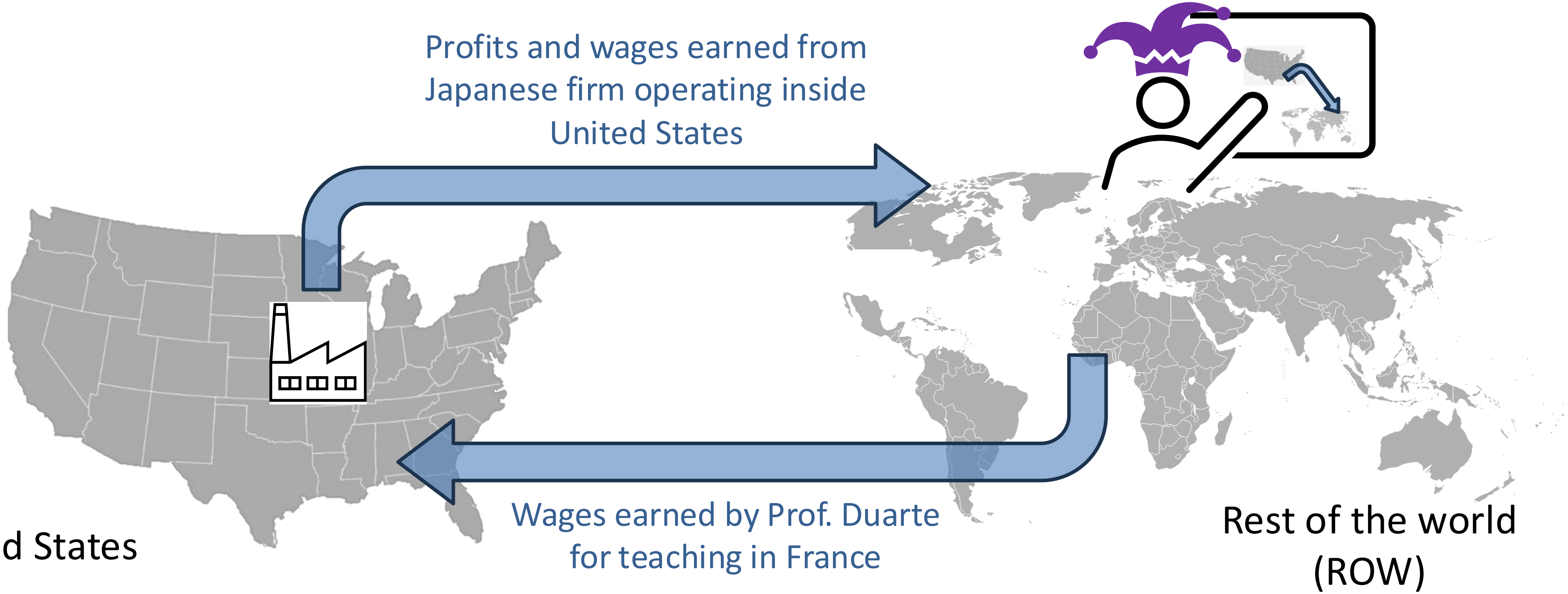
$$\text{GNP} = \text{GDP} + \text{Net Income from Abroad}$$

Open Economy with GNP

$$GNP = C + I + G + CA$$

- CA = Current account
- $CA = NX +$ Net income from abroad

GDP and GNP



Net Income = wages from Prof. Duarte teaching in France − profits and wages from firm operating inside US

United States GNP = United States GDP + Net Income

National Saving: Closed Economy

National saving = output not used for consumption or government spending

$$\begin{aligned} S &\equiv Y - C - G \\ &= (C + I + G) - C - G \\ &= I \end{aligned}$$

In a closed economy: $S = I$

National Saving: Open Economy

Starting from $Y = C + I + G + NX$:

$$S \equiv Y - C - G$$

$$= I + NX$$

In an open economy: $S = I + NX$

National Saving: Open Economy

Rearranging: $S - I = NX$

- If $S > I$: Trade surplus, country is a net lender
- If $S < I$: Trade deficit, country is a net borrower

Private Saving

Private saving = disposable income that is saved, not consumed

$$S^p \equiv Y - T - C$$

Public (Government) Saving

Government saving = tax revenue minus government spending

$$S^g \equiv T - G$$

- If $T > G$: Budget surplus ($S^g > 0$)
- If $T < G$: Budget deficit ($S^g < 0$)

Total Saving

Total national saving:

$$\begin{aligned} S &= S^p + S^g \\ &= (Y - T - C) + (T - G) \\ &= Y - C - G \end{aligned}$$

Using $S = I + NX$ from before, we get:

$$S^p + S^g = I + NX$$

Balance of Payments

Current Account	-200
Capital Account	-1
Financial Account	-201

CA = NX + NI

NON-MARKET TRANSACTIONS

FINANCIAL FLOWS

$$\text{Current Account} + \text{Capital Account} = \text{Financial Account}$$

$$(-200) + (-1) = -201$$

Balance of Payments

Current Account	-200	-201
Capital Account	-1	
Financial Account	-201	-201
Statistical discrepancy	$\underline{0}$	"MEASUREMENT ERROR"

$$\begin{array}{ccccccccc} \text{Statistical Discrepancy} & = & \text{Financial Account} & - & \Big(& \text{Current Account} & + & \text{Capital Account} & \Big) \\ \underline{0} & = & -201 & - & [& (-200) & + & (-1) &] \end{array}$$

U.S. Balance of Payments Accounts for 2025-Q3	(\$ bn)
Current Account	-226
Capital Account	-1
Financial Account	-410
Statistical discrepancy	-183

$$\begin{array}{ccccccccc} \text{Statistical Discrepancy} & = & \text{Financial Account} & - & \left(\text{Current Account} \right. & + & \left. \text{Capital Account} \right) \\ -183 & = & -410 & - & [\ (-226) & + & (-1)\ ] \end{array}$$

Source: BEA

Current Account

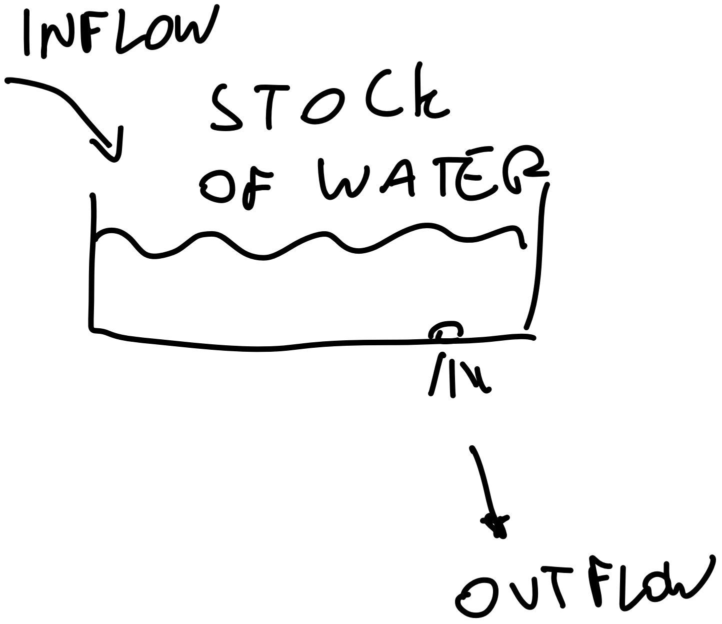
1	Exports total	
2	Goods	
3	Services	
4	Income receipts (primary income)	PART OF NET INCOME → LIKE SERVICES PROVIDED BY AMERICAN FACTORS TO ROW
5	Imports total	
6	Goods	
7	Services	
8	Income payments (primary income)	PART OF NET INCOME
9	Net unilateral transfers (secondary income)	
10	**Balance on current account**	

Capital Account

11	Balance on capital account	

Financial Account

12	Net U.S. acquisition of foreign financial assets	
13	✓ Official reserve assets	→ CENTRAL BANKS ✓
14	Other assets	→ PRIVATE SECTOR
15	Net U.S. incurrence of domestic liabilities	
16	Official reserve assets	
17	Other liabilities	
18	Financial derivatives, net	
→ 19	**Net financial flows**	(12) − (15) + (18)
20	**Statistical discrepancy**	



Bottom line to remember

Whenever goods cross borders, assets move in the opposite direction.